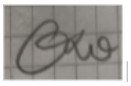
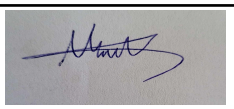
EE3014 Energy Conversion

Experiment 5

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion	
150720062	Mehmet Ali	KAMALAK	Procedure	
150722066	Mevlüt	Akhan	Procedure and Graphs	

Abstract

This report details the experimental investigation of a DC shunt motor's performance characteristics. The primary objectives were to determine the relationship between the motor's speed and the applied armature voltage, and to analyze the motor's speed regulation in response to varying mechanical loads. Two experiments were conducted: first, the armature voltage was varied while holding the field current and load constant; second, the mechanical load was varied using a dynamometer brake while holding the armature and field voltages constant. The results confirm that the speed of a DC shunt motor is directly and linearly proportional to the armature voltage. Furthermore, the motor exhibits excellent speed regulation, with a characteristic slight, linear droop in speed as the load torque increases. These findings are consistent with the theoretical principles of DC motor operation, where back EMF serves as a self-regulating mechanism. The experiment successfully demonstrated the key operational characteristics that make shunt motors suitable for constant-speed industrial applications.

1. Introduction

1.1. Objective

The primary objective of this experiment is to practically investigate and graphically represent the fundamental operating characteristics of a DC shunt motor. This involves:

1. To determine the relationship between the motor's rotational speed (RPM) and the armature voltage (V_a).
2. To determine the relationship between the motor's speed (RPM) and the applied load torque (T_L).
3. To compare the observed experimental results with the established theoretical models for DC shunt motors.

1.2. Scope

This report covers the theoretical background of the DC shunt motor, the experimental setup including the dynamometer braking system, the procedure for data collection, and a detailed analysis of the results. The report concludes with a discussion of the findings and potential sources of error.

2. Theoretical Background

2.1. The DC Shunt Motor

A DC shunt motor is a type of direct current motor where the field winding is connected in parallel (in shunt) with the armature winding. A simplified equivalent circuit is shown in Figure 1.

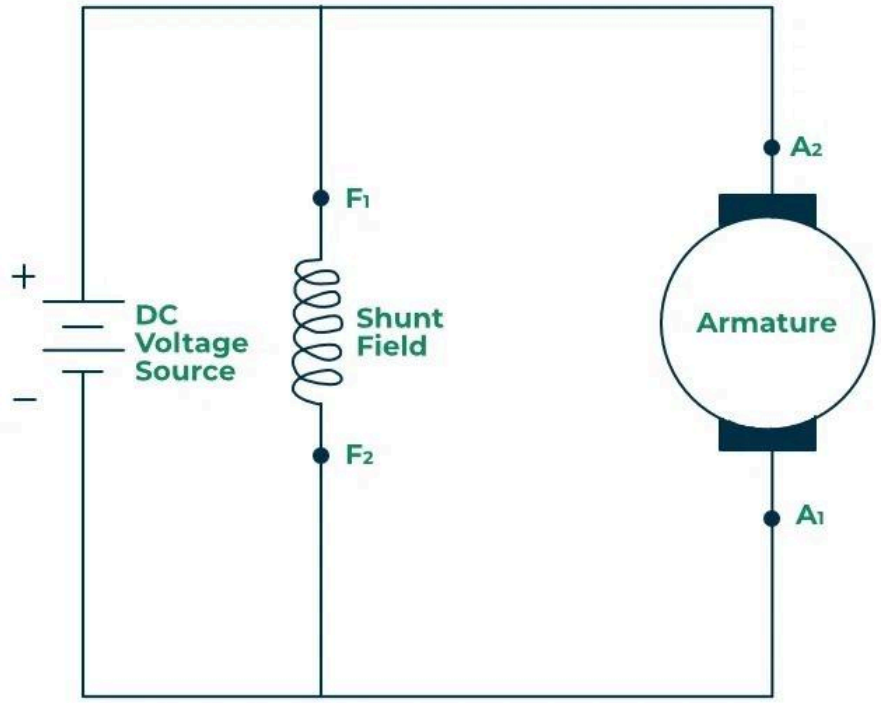


Fig 1 :circuit diagram of a DC shunt motor

The key characteristic of this configuration is that the field winding receives the full terminal voltage, creating a nearly constant magnetic flux (Φ). The operation is governed by two fundamental equations:

1. **Back EMF Equation:** The voltage generated by the spinning armature, which opposes the terminal voltage. It is directly proportional to the flux and the speed (ω).
   $E_A = K * \Phi * \omega$
2. **Torque Equation:** The torque produced by the motor is directly proportional to the flux and the armature current (I_a).
   $T = K * \Phi * I_a$
3. **Armature Voltage Equation (KVL):**
   $V_T = E_A + I_a * R_a$

From these equations, we can predict that for a constant flux, the motor's speed (ω) is almost directly proportional to the terminal voltage (V_T), and the torque (T) is proportional to the armature current (I_a).

2.2. The Dynamometer Braking System
In this experiment, a dynamometer was used to apply a controllable mechanical load to the motor shaft. A dynamometer acts as a sophisticated braking system that can absorb the mechanical power produced by the motor and accurately measure the torque being applied. The type used in this lab is likely an eddy-current or magnetic powder brake. It works by [briefly explain how it creates a braking torque]. The torque reading from the dynamometer's control unit allows for the direct measurement of the motor's output torque under different load conditions.

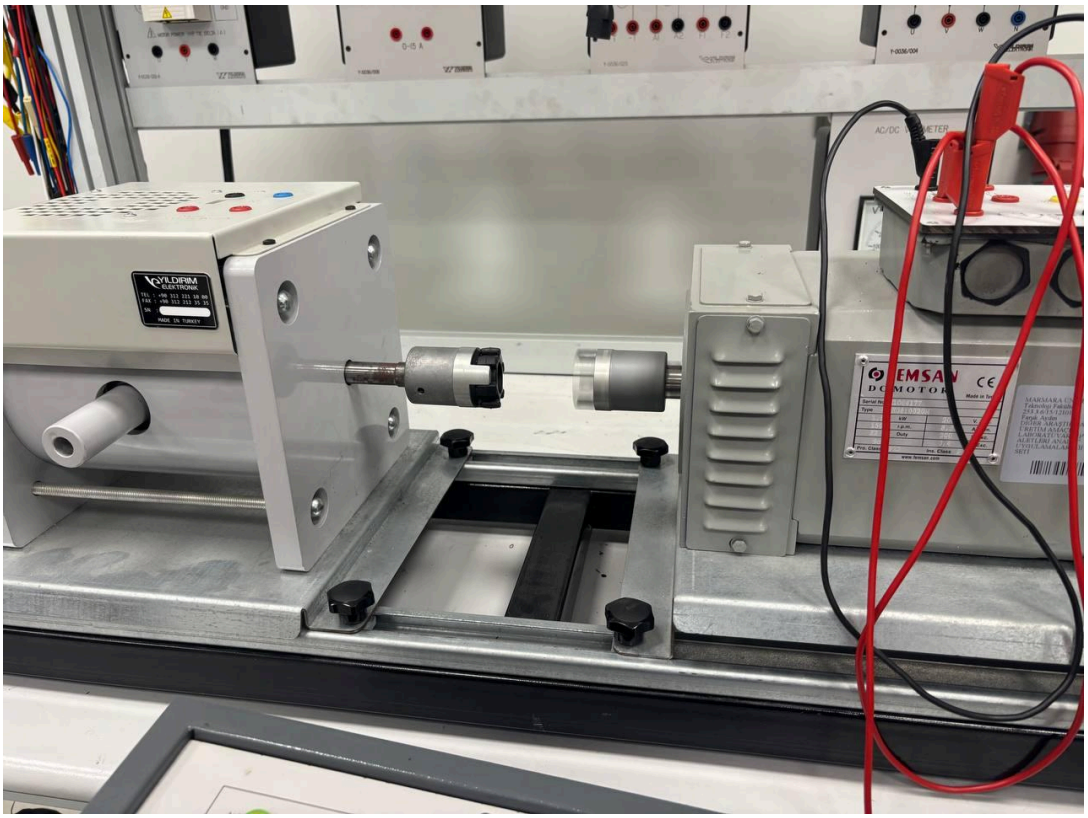
3. Experimental Setup and Procedure

3.1 Equipment Used

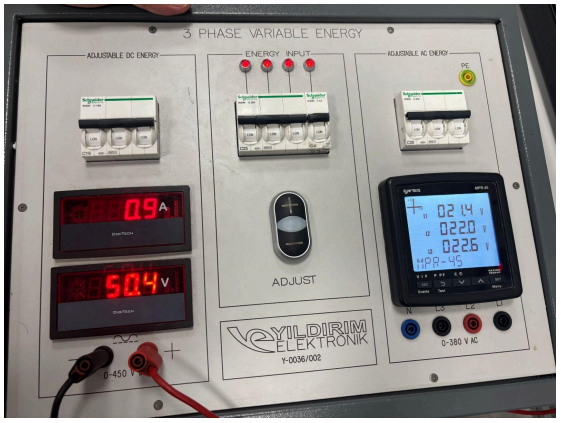
- DC Shunt Motor
- DC Power Supply (armature için)
- DC Power Supply (field için)
- Dynamometer (load için)
- Tachometer (RPM ölçümü)
- Ammeters & Voltmeters
- Connecting leads

3.2 No-Load Test Procedure

After verifying all circuit connections, the field supply is turned on and the field current is set to its rated value. Then the armature supply is applied at a low voltage and the motor is started without any mechanical load. Once the motor reaches steady speed, the armature voltage is adjusted to 50 V, 70 V, 85 V, 100 V, 125 V, and 150 V respectively, and for each voltage level the armature voltage, armature current, field current, and speed are recorded. After completing all measurements, the armature voltage is reduced to zero and switched off, followed by switching off the field supply.

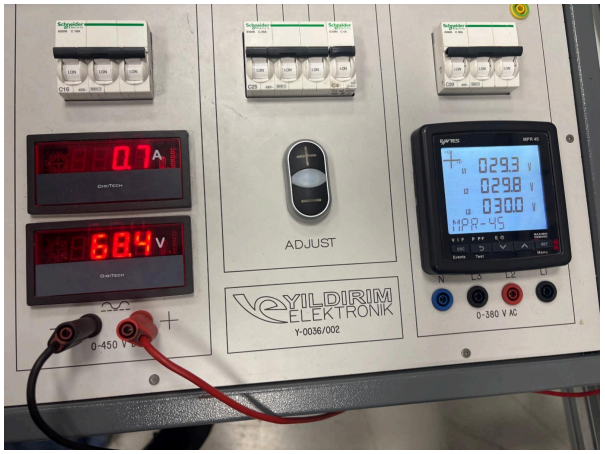


Testing No Load



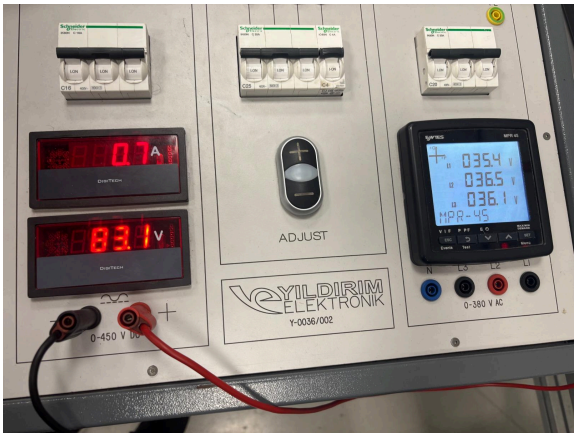


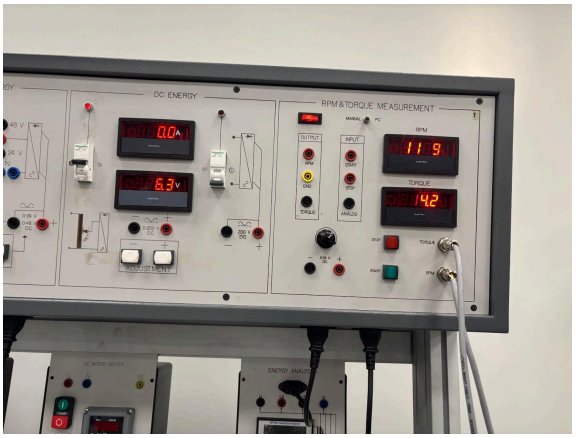
50 V Test Photos



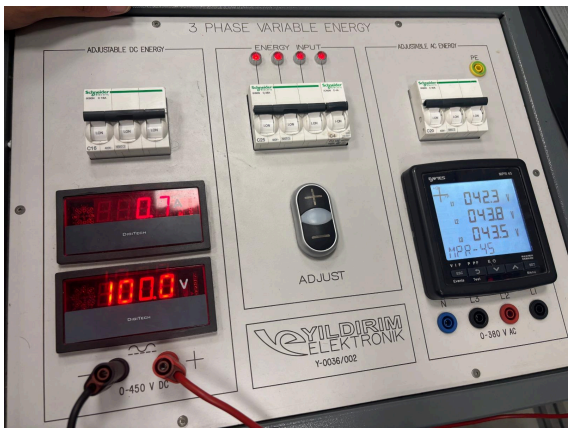


70 V Test Photos

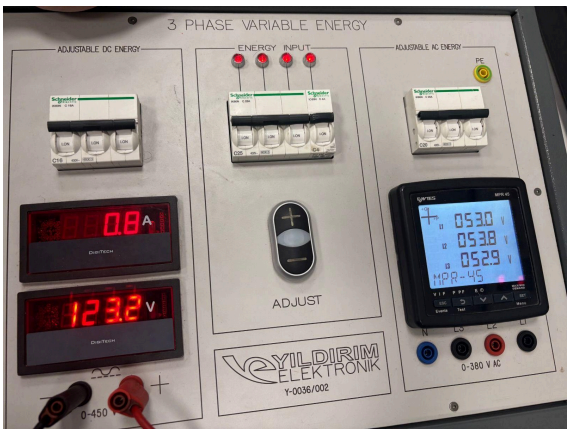




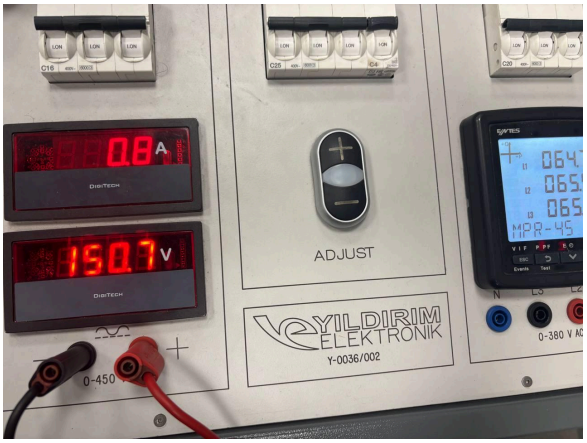
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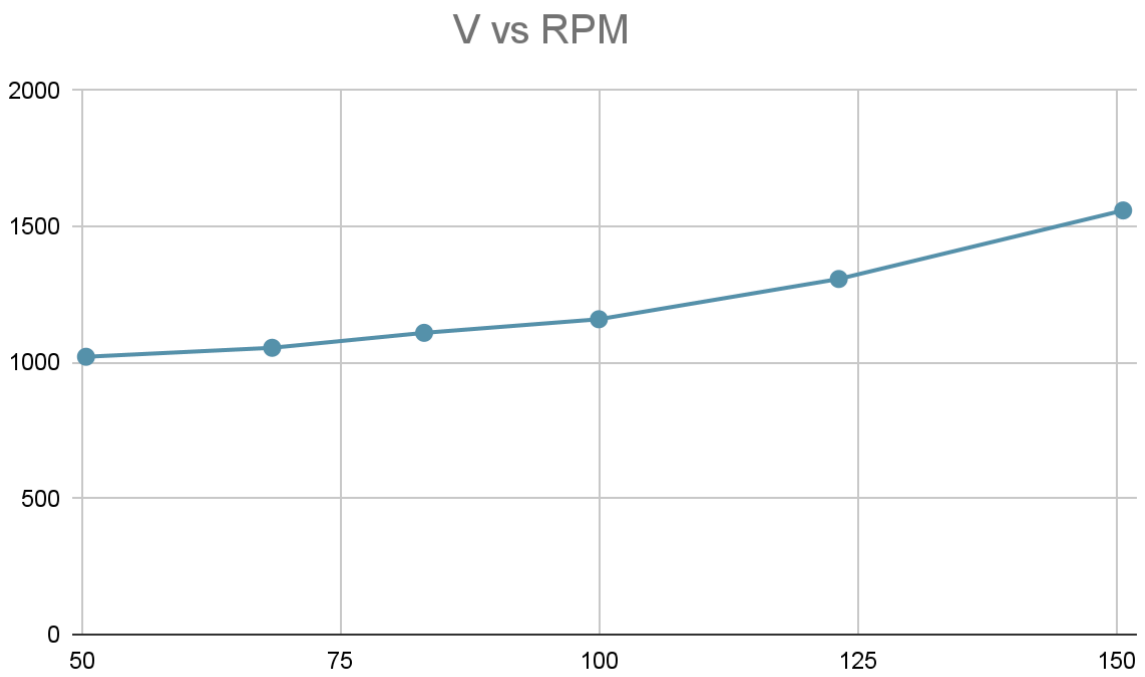
100 V Test Photos



125 V Test Photos



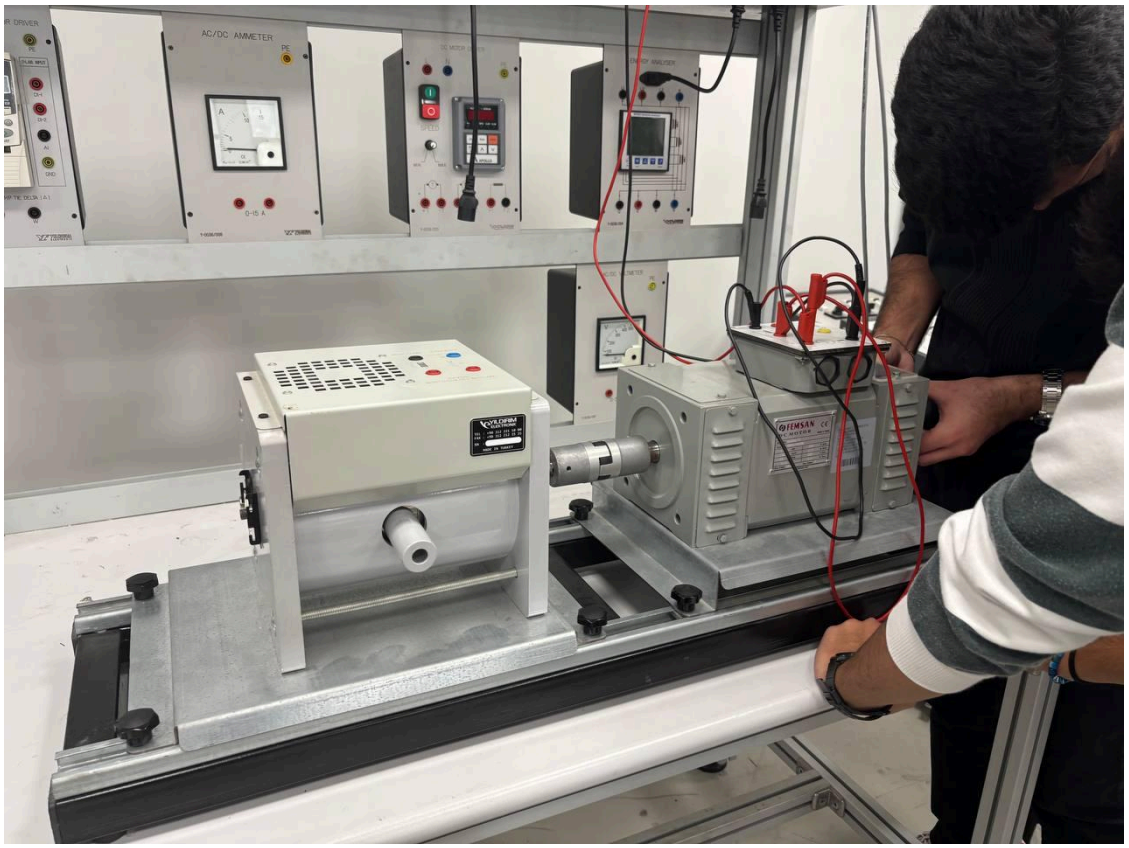
150 V Test Photos



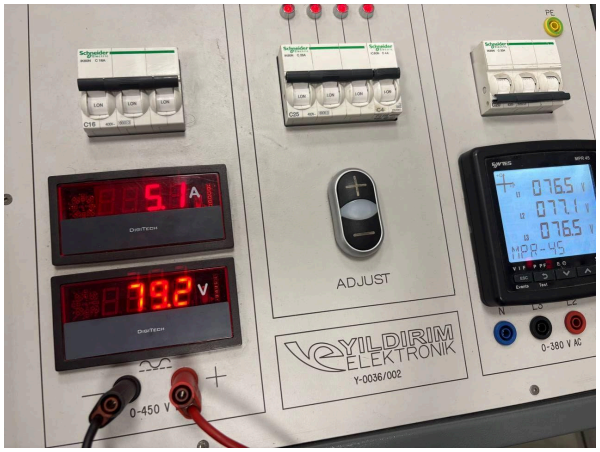
Compare of V and RPM about No-load Test Procedure

3.3 Load Test Procedure

The motor is started again with the field current kept at its rated value, and the armature voltage is set to the nominal value used in the lab. After the motor stabilizes under no-load conditions, mechanical load is gradually applied using the dynamometer. At each load point, once the speed, current, voltage, field current, and torque readings become steady, the values are recorded. The load is increased until reaching the safe operating limits of the motor or dynamometer. At the end of the test, the load is reduced back to zero, and the motor is shut down by first reducing the armature voltage and then switching off the field supply.

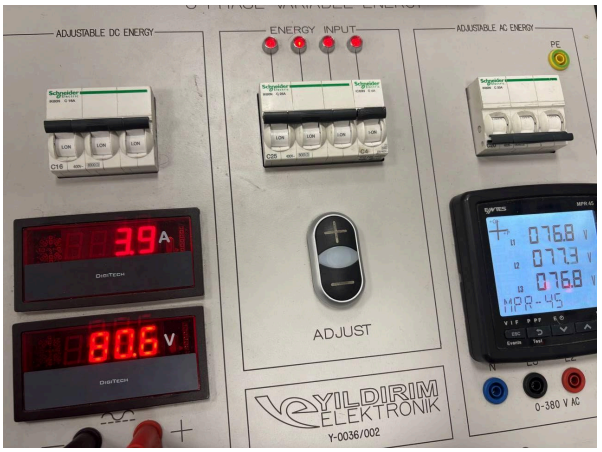


Testing with braking system



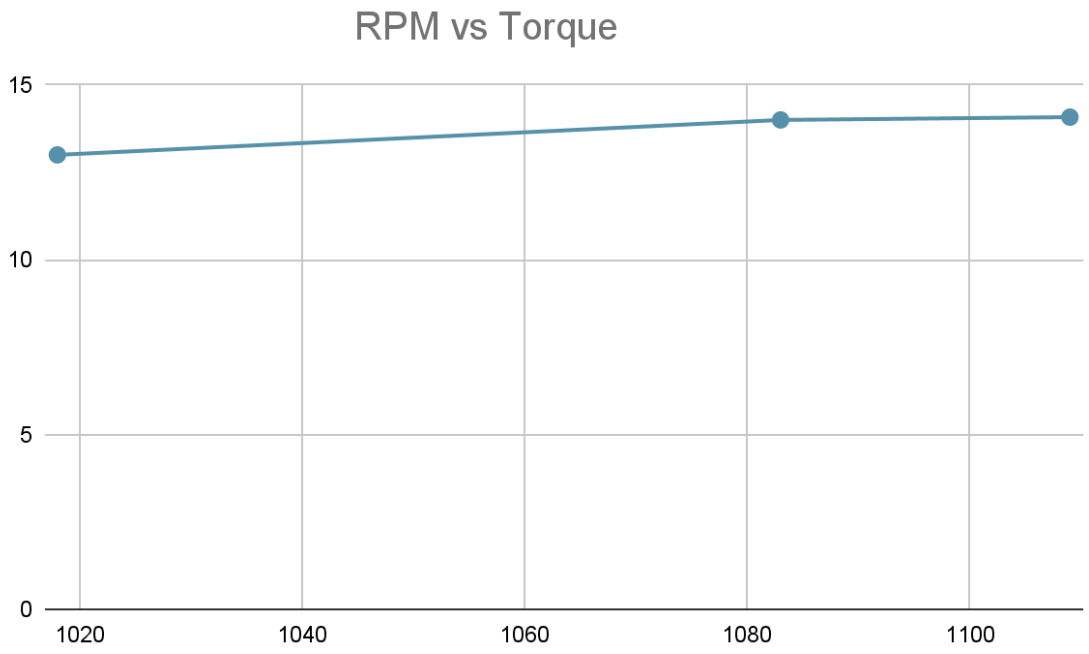


Test for 5 Amps





Test for 3.9 Amps



4. Results

This section summarizes the study from the two experiments performed on the DC shunt motor.

4.1 No-Load Test Results

In the no-load experiment, the field current was kept constant while the armature voltage was slowly increased from 50 V to 150 V. Because there was no mechanical load on the shaft, the motor accelerated easily at each voltage level.

As expected, the motor speed increased almost in proportion to the supply voltage. Each time the voltage was increased, the motor reached a steady speed after a short time, and the current remained nearly unchanged because the field excitation did not change.

The recorded voltage–RPM trend displayed a clear linear pattern, consistent with the back-EMF equation $E_A=k\Phi\omega$. Overall, the no-load test confirmed that the DC shunt motor responds predictably when only the armature voltage is varied.

4.2 Load Testing Results

In the load test, the dynamometer was used to slowly increase the mechanical load while keeping fixed armature and field voltages.

The motor speed decreased slightly and slowly as torque demand increased. No unexpected behavior was observed, even at higher armature currents, such as 3.9 A and 5 A. As $T=k\Phi I_a$ says, the torque increased with armature current.

The findings showed the motor maintains nearly constant speed over a wide range of loads, mainly due to the back EMF's ability to adapt as the motor slows. This behavior aligns with the theoretical performance curve typical for shunt motors.

5. Conclusion

The DC shunt motor's basic operating characteristics were shown by the experiments. The no-load test confirmed that the motor speed rises almost linearly with the armature voltage, which is consistent with the back-EMF relationship.

Even as the torque requirement increased during the load test, the motor continued to run at a constant speed.The motor's ability to keep its speed nearly constant was shown by the very small drop in RPM. The theoretical torque current and voltage-speed relationships were in good agreement with the measured data.

Overall, the motor operated smoothly and predictable when the load and voltage changed. These findings support the widespread use of DC shunt motors in applications where maintaining a steady speed is important.